MIL-R-23417A
27 June 1975
SUPERSEDING
MIL-R-23417
13 August 1963

MILITARY SPECIFICATION

RESOLVERS, ELECTRICAL: GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification relates to electrical resolvers. It is not complete in itself but shall be used in conjunction with MIL-S-81963 in which the latter shall be recognized as forming an inherent part of this specification. (See 6.2).

1.2 Classification.

1.2.1 Nomenclature.- The nomenclature shall consist of the item name, i.e., Resolver, Electrical followed by a type designation and a part number (see 6.2). All resolvers having the same nomenclature shall be mechanically and electrically interchangeable for all military applications. The type designation shall be indicated by a combination of digits and letters. Illustrated below is the complete nomenclature for a resolver type 23R32N4B.

ITEM NAME	23	R	32	N	4	B	M23417/16-01
1.2.1	1.2.1.1	1.2.1.2	1.2.1.3	1.2.1.4	1.2.1.5	1.2.1.6	1.2.1.7

1.2.1.1 Size.- The first two numerals indicate the maximum external diameter of the resolver in tenths of an inch. If the diameter is not exactly a whole number of tenths, the next higher tenth is used, e.g., "23" indicates a resolver with a maximum body diameter of 2.250 inches.

1.2.1.2 Function.- The letter "R" after "Size" designates resolver.

1.2.1.3 Impedance of primary winding.- The following numerals after "Function" indicate the nominal impedance of the primary winding in hundreds of ohms. If the impedance is not exactly a whole number of hundreds, the next higher hundred is used.

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1.2.1.4 Compensation.- After "Impedance of primary winding", compensation is indicated as follows:

Winding compensated	W
Not compensated	N

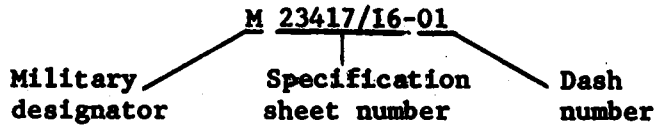
1.2.1.5 Rated frequency.- The compensation denoting letter is followed by numerals indicating the rated frequency of the system power supply from which the resolver is designed to operate.

Rated Frequency (Hz)	Code
60	6
400	4
800	8
1000	10
10000	100
20000	200

NOTE: The designation of the resolver by the frequency code "6" does not imply that 60 Hz must necessarily be used. It will generally be possible to employ a 50 Hz supply with a device to reduce the voltage appropriately, although certain of the resolver characteristics will thereby be modified.

1.2.1.6 Modification.- The upper case letter "A" following the frequency digit indicates the original or basic design standard of the resolver type. The first modification to affect the physical interchangeability or the performance characteristics of the basic type is indicated by the upper case letter "B", and succeeding modifications are indicated by "C", "D" etc.

1.2.1.7 Military part number.- The military part number following the modification letter shall consist of the letter "M", the specification sheet number, and an assigned dash number (see 3.1) as illustrated:



2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MIL-S-81963 Servo-Components, Precision Instrument, Rotating, Common Requirements and Tests: General Specification For

(See Supplement 1 for list of applicable specification sheets.)

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Specification sheets.- The individual item requirements shall be as specified herein, MIL-S-81963, and in accordance with the applicable specification sheets. (See 6.2 and 6.7).

3.2 Qualification.- Resolvers furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.7 and 6.5).

3.3 First article.- When required by the procuring activity, resolvers shall conform to the requirements specified herein and shall have met the first article inspection specified in 4.8 prior to regular production on a contract (see 6.3).

3.4 Design conventions.

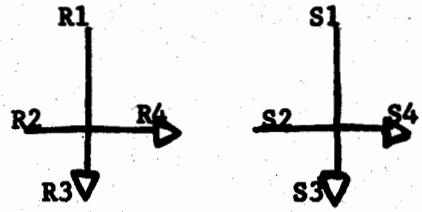
3.4.1 Direction of rotation.- The standard (positive) direction of rotation of the shaft is counter-clockwise when the resolver is viewed from the shaft extension end.

3.4.2 Electrical angle.- The electrical angle is the angle θ in the equations which define the ideal magnitude and polarity relationships among secondary voltages. Defining equations for any rotor angle (see 6.4.1) and diagrams of relative instantaneous polarity at resolver zero are as follows:

3.4.2.1 Rotor energized resolvers.

$$E(S1S3)=NE(R1R3)\cos\theta - NE(R2R4)\sin\theta$$
$$E(S2S4)=NE(R2R4)\cos\theta + NE(R1R3)\sin\theta$$

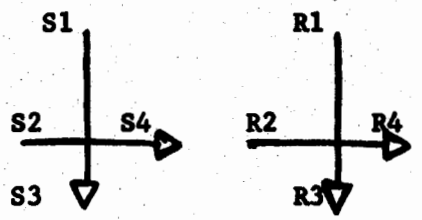
where N = transformation ratio



3.4.2.2 Stator energized resolvers.

$$E(R1R3)=NE(S1S3)\cos\theta + NE(S2S4)\sin\theta$$
$$E(R2R4)=NE(S2S4)\cos\theta - NE(S1S3)\sin\theta$$

where N = transformation ratio



3.4.3 Resolver zero.- Resolver zero is the position of the rotor shaft for which the electrical angle is zero. The electrical angle is zero when the position of the rotor with respect to the stator satisfies the secondary voltage polarity relationships of 3.4.2 for the resolver zero position, and minimum voltage is induced in the specified secondary winding. The minimum

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voltage occurs at the rotor position where the fundamental frequency component is zero and the secondary voltage at that point is in time phase with the maximum value of secondary voltage for the same winding. The pertinent maximum value of secondary voltage is the first maximum position encountered when the rotor is turned in the positive direction from resolver zero.

3.4.3.1 Rotor energized resolvers.- Resolver zero occurs for rotor energized resolvers when windings R1R3 and S2S4 are at minimum coupling and is determined in accordance with Figure 1.

3.4.3.2 Stator energized resolvers.- Resolver zero occurs for stator energized resolvers when windings S1S3 and R2R4 are at minimum coupling and is determined in accordance with Figure 2.

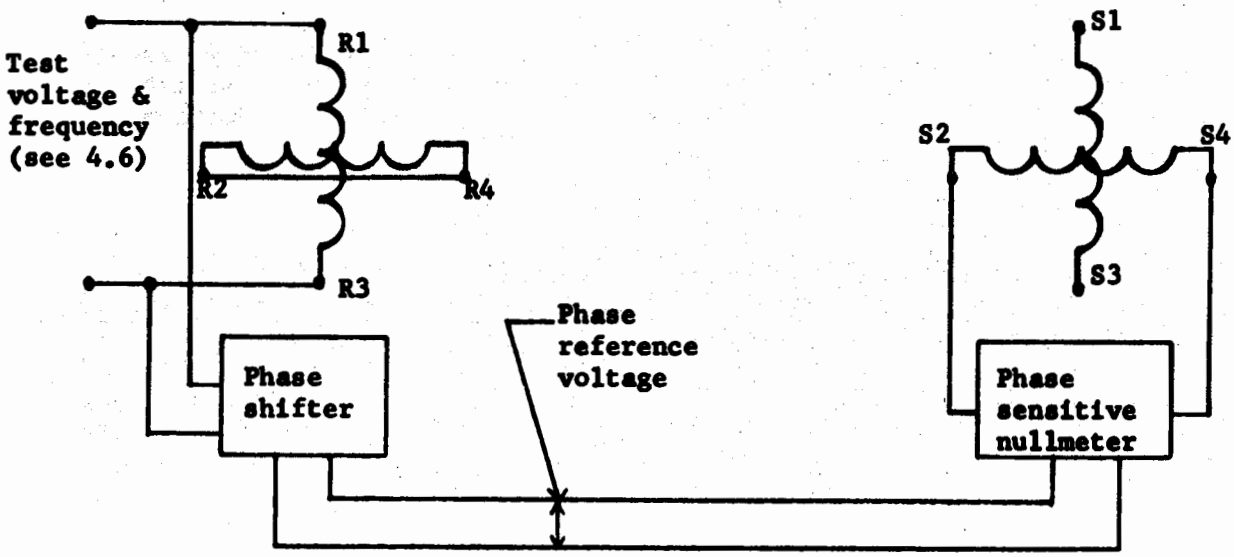


Figure 1. Resolver Zero, Rotor Energized Resolvers.

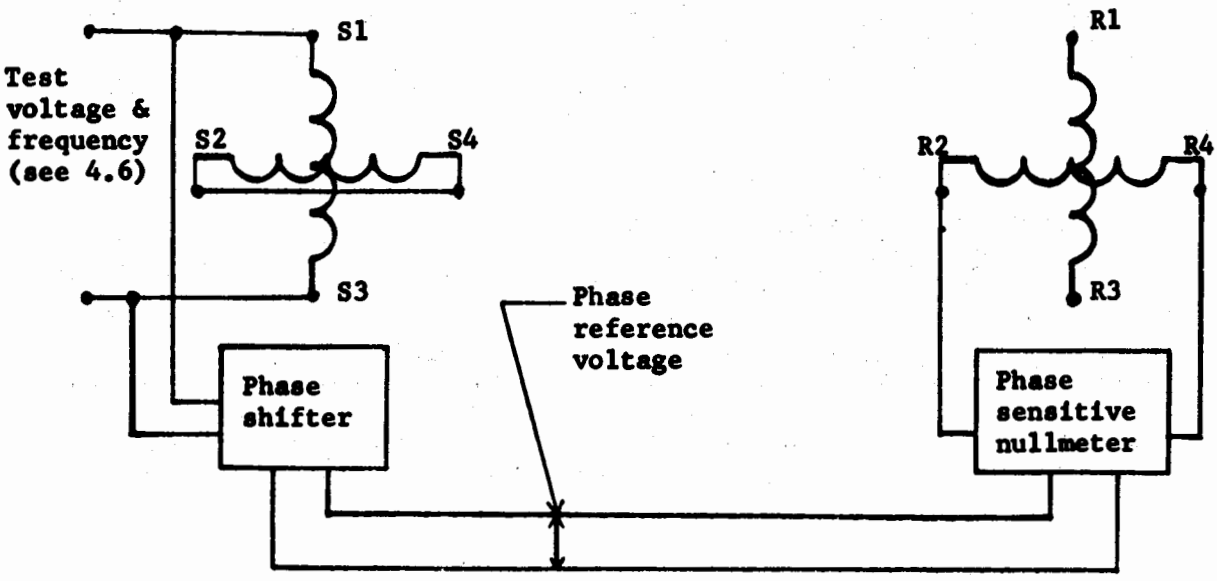


Figure 2. Resolver Zero, Stator Energized Resolvers.

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3.5 Design and construction.

3.5.1 Terminals and wire leads.- Winding terminations shall be as specified in the applicable specification sheet. Constructional requirements of screw thread and solder pin terminal types and wire lead types shall conform to MIL-S-81963. Termination identification marking for screw thread and solder pin terminal types shall be in accordance with Table I. Wire lead types shall be color coded in accordance with Table I.

TABLE I Termination Identification Marking

Terminal No.	Wire Lead Color	Winding
R1	Red, White Tracer	Rotor, Start
R3	Black, White Tracer	Rotor, Finish
R2	Yellow, White Tracer	Rotor, Start
R4	Blue, White Tracer	Rotor, Finish
S1	Red	Stator, Start
S3	Black	Stator, Finish
S2	Yellow	Stator, Start
S4	Blue	Stator, Finish
C1	Red, Green Tracer	Compensating, Start
C3	Black, Green Tracer	Compensating, Finish
C2	Yellow, Green Tracer	Compensating, Start
C4	Blue, Green Tracer	Compensating, Finish

NOTE: Terminal designations are such that the stator and compensating terminals with like numbers have like polarity when the resolver rotor angle is zero degrees.

3.5.2 Resolver zero marking.- Resolver zero marking shall be in accordance with MIL-S-81963 and the applicable specification sheet.

3.6 Performance.

3.6.1 Variation of brush contact resistance.- The variation of brush contact resistance between winding terminations R2R4 and R1R3 of the resolver shall be as specified in MIL-S-81963.

3.6.2 Shaft radial and end play.- Shaft radial and end play of the resolver rotor shaft shall be in accordance with MIL-S-81963 and the specified limits of the applicable specification sheet.

3.6.3 Mechanical breakaway torque.- Mechanical breakaway torque shall be in accordance with MIL-S-81963 and the torque required to turn the resolver rotor shaft shall not exceed the value specified in the applicable specification sheet.

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3.6.4 Dielectric withstanding voltage.- The resolver shall withstand a dielectric stress voltage in accordance with MIL-S-81963 and the resolver shall show no evidence of failure as specified therein. The voltages and application points shall be in accordance with Table III.

3.6.5 Insulation resistance.- Unless otherwise specified in the applicable specification sheet the insulation resistance between the application points designated in Table III shall be measured in accordance with MIL-S-81963 and shall not be less than the values specified therein. For resolvers with maximum operating voltages greater than 50 volts rms, 100 volts dc shall be applied between application points involving windings in intimate contact and 500 volts dc shall be applied between all other application points. For resolvers with maximum operating voltages less than 50 volts rms, 100 volts dc shall be applied between all application points.

3.6.6 Current.- The current drawn by each primary winding shall be in accordance with MIL-S-81963 and shall be within the limits specified in the applicable specification sheet.

3.6.7 Power.- The power consumption of each primary winding shall be in accordance with MIL-S-81963 and shall be within the limits specified in the applicable specification sheet.

3.6.8 Transformation ratio. (See 6.4.13)

3.6.8.1 Transformation ratio.- The transformation ratio for each rotor/stator, stator/rotor or rotor/compensator winding combinations shall be within the limits specified in the applicable specification sheet.

3.6.8.2 Transformation ratio variation with voltage.- The transformation ratio for each winding combinations with maximum and minimum operating voltages shall be within the limits specified in the applicable specification sheet when compared to the value obtained at the test voltage.

3.6.8.3 Equality of transformation ratios.- The maximum difference between the transformation ratios measured in 3.6.8.1 shall be within the limits specified in the applicable specification sheet.

3.6.9 Phase shift. (See 6.4.8)

3.6.9.1 Phase shift.- The phase shift for each rotor/stator, stator/rotor or rotor/compensator winding combination shall be within the limits specified in the applicable specification sheet.

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3.6.9.2 Phase shift variation with voltage.- The phase shift variation for each winding combination at maximum and minimum operating voltages shall not exceed the limits specified in the applicable specification sheet.

3.6.10 Function error. (See 6.4.3).- Function error shall not exceed the value specified in the applicable specification sheet.

3.6.11 Electrical error. (See 6.4.2).- Electrical error shall not exceed the value specified in the applicable specification sheet.

3.6.12 Residual (null) voltage. (See 6.4.10).- Fundamental and total residual voltages shall be within the limits specified in the applicable specification sheet. For qualification, the resolver shall also meet this requirement at maximum operating voltage as specified in the applicable specification sheet.

3.6.13 Interaxis error. (See 6.4.7).- Interaxis error shall not exceed the value specified in the applicable specification sheet.

3.6.14 Impedance. (See 6.4.5).- Rotor, stator, and where applicable, compensating winding impedances shall be in accordance with MIL-S-81963 and shall be within the limits specified in the applicable specification sheet.

3.6.15 Compensating windings.

3.6.15.1 Residual voltage, compensating windings.- Compensating winding fundamental residual voltage shall be within the limits specified in the applicable specification sheet.

3.6.15.2 Winding phase, compensating windings.- Like numbered terminations of stator and compensating windings shall be of the same polarity when the resolver is energized at its test voltage.

3.6.15.3 Equality of compensating windings.- The maximum difference between fundamental residual voltages of compensating windings shall be within the limits specified in the applicable specification sheet.

3.6.16 Security of terminals or wire leads.- The security of each screw type or solder pin type of terminal or of each wire lead, as applicable to the particular type of resolver shall be in accordance with MIL-S-81963 and shall meet the requirements specified therein.

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3.6.17 Harmonic distortion. (See 6.4.4).- Harmonic distortion shall not exceed the value specified in the applicable specification sheet.

3.6.18 Shift of resolver zero.

3.6.18.1 Shift of resolver zero with voltage.- The change of position of resolver zero with variation of maximum and minimum primary voltage shall be within the limits as specified in the applicable specification sheet.

3.6.18.2 Shift of resolver zero with frequency.- The change of position of resolver zero when the rated frequency is first decreased by 10 percent and increased by 10 percent shall be within the limits as specified in the applicable specification sheet.

3.6.19 Resonant frequency response.- The resolver shall meet the minimum requirement for resonant frequency response for each primary winding combination as specified in the applicable specification sheet.

3.6.20 Temperature rise.- The temperature rise calculated from the expression in MIL-S-81963 shall not exceed the value specified in the applicable specification sheet.

3.6.21 Electromagnetic interference.- Electromagnetic conducted and radiated interference of the resolver shall be in accordance with and shall not exceed the limits of MIL-S-81963. Both resolver primary windings shall be energized and the rotor shall be rotated at 600 $\pm$ 50 rpm. A load equivalent to four times the secondary winding open circuit impedance shall be applied across both secondary windings. Compensating windings shall be unloaded.

3.7 Environmental.

3.7.1 Vibration.- The resolvers shall meet the requirements as specified in MIL-S-81963 in addition to the requirements of Table II as listed herein.

3.7.2 Shock.

3.7.2.1 Shock, specified pulse.- The resolvers shall meet the requirements as specified in MIL-S-81963 in addition to the requirements of Table II as listed herein.

3.7.2.2 Shock, high impact.- The resolvers shall meet the requirements as specified in MIL-S-81963 in addition to the requirements of Table II as listed herein.

3.7.3 Altitude.

3.7.3.1 Altitude, low temperature.- The resolvers shall meet the requirements as specified in MIL-S-81963 in addition to the requirements of Table II as listed herein.

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3.7.3.2 Altitude, high temperature.- The resolvers shall meet the requirements as specified in MIL-S-81963 in addition to the requirements of Table II as listed herein.

3.7.4 Endurance.- The resolvers shall operate for a minimum of 2000 hours in accordance with the requirements of MIL-S-81963 while energized and while the rotor is being rotated at 1150 ± 50 rpm. Immediately after meeting this requirement, the resolvers shall meet the requirements of Table II as listed herein.

3.7.5 Ambient temperature.

3.7.5.1 Ambient low temperature.- The resolvers shall operate as specified in MIL-S-81963 and shall meet the requirements of Table II as listed herein.

3.7.5.2 Ambient high temperature.- The resolvers shall operate as specified in MIL-S-81963 and shall meet the requirements of Table II as listed herein.

3.7.6 Moisture resistance.- The resolvers shall operate as specified in MIL-S-81963 and shall meet the requirements of Table II as listed herein.

3.7.7 Explosion.- When specified by the applicable specification sheet, the resolvers shall operate without causing an explosion as specified in MIL-S-81963.

3.7.8 Salt atmosphere.- When specified by the applicable specification sheet, the resolvers shall withstand atmosphere saturated with salt-laden moisture as specified in MIL-S-81963.

3.7.9 Identification marking.- The resolvers shall meet the identification marking requirement as specified in MIL-S-81963. The nomenclature to be employed shall be in accordance with 1.2.1 and 3.1.

3.8 Workmanship.- The resolver shall meet the workmanship requirement of MIL-S-81963.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 General.- Unless otherwise specified herein or in the applicable specification sheet all testing shall be conducted in accordance with MIL-S-81963.

4.3 Test conditions.- Unless otherwise specified each test shall be carried out with the resolver mounted in a standard test fixture applicable to its frame size in accordance with MIL-S-81963. When approved by the qualifying activity, an equivalent induction type resolver bridge or resolver standard, where applicable, may be used for electrical error testing.

4.3.1 Angular accuracy test stand.- When specified, the angular accuracy test stand shall be an angular dividing device by which angular positions may be set with an error no greater than ± 15 seconds of arc. The means adopted to mount the resolver shall be such that accumulated tolerances do not introduce an error due to lack of concentricity greater than 30 seconds of arc.

4.4 Temperature, stabilized, non-operating.- The stabilized non-operating temperature of the resolver shall be as specified in MIL-S-81963 using a secondary winding for the periodic dc resistance measurement.

4.5 Temperature, stabilized, operating.- The stabilized operating temperature of the resolver shall be as specified in MIL-S-81963 using a secondary winding for the periodic dc resistance measurement.

4.6 Test voltage and frequencies.- The test voltage and frequency of the resolver are those specified for the primary winding in the applicable specification sheet. Performance values and their limits stated in the applicable specification sheets are based on ± 1 percent tolerance on voltage and frequency, on 1 percent harmonic content, and on test conditions which are closely controlled. Unless otherwise specified the amplitude and frequency of test voltages shall be within 1 percent of the rated or specified value. The total harmonic content of test voltages shall not exceed 1 percent of the amplitude of the voltage of fundamental frequency. In addition, at all corresponding ordinates, the divergence of the waveform of the test voltage from that of a pure sine wave of the same rms value shall not exceed 1 percent of the instantaneous value of the sine wave.

4.7 Qualification inspection.- Qualification inspection shall be performed at a laboratory approved by the US Army Armament Command (see 6.5). Qualification inspection shall be in accordance with MIL-S-81963 and Table II herein.

4.8 First article inspection.- When required by the procuring activity (see 6.2), first article inspection shall be in accordance with MIL-S-81963 and 4.7 herein as applicable.

4.9 Quality conformance inspection.- Quality conformance inspection shall be in accordance with MIL-S-81963 and Table II herein.

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TABLE II - QUALIFICATION AND QUALITY CONFORMANCE INSPECTION

TEST NO.	REQ'T	TEST	TEST METHOD OR EXAMINATION	INSPECTION	
				QUALIFICATION Sample Nos.	QUALITY CONFORMANCE
1	3.6.1	4.10.1	Variation of brush contact resistance	1, 2, 3, 4	X
2	3.1 & 3.8	4.10.2	Visual and mechanical examination	1, 2, 3, 4	X
3	3.6.2	4.10.3	Shaft radial and end play	1, 2, 3, 4	X
4	3.6.3	4.10.4	Mechanical breakaway torque	1, 2, 3, 4	X
5	3.6.4	4.10.5	Dielectric withstanding voltage	1, 2, 3, 4	X
6	3.6.5	4.10.6	Insulation resistance	1, 2, 3, 4	X
7	3.6.6	4.10.7	Current	1, 2, 3, 4	X
8	3.6.7	4.10.8	Power	1, 2, 3, 4	X
9	3.6.8.1	4.10.9.1	Transformation ratio	1, 2, 3, 4	X
10	3.6.8.3	4.10.9.3	Equal. of transformation ratios	1, 2, 3, 4	X
11	3.5.2	4.10.20	Resolver zero marking	1, 2, 3, 4	X
12	3.6.9.1	4.10.10.1	Phase shift	1, 2, 3, 4	X
13	3.6.10	4.10.11	Function error 1/	1, 2, 3, 4	X
14	3.6.11	4.10.12	Electrical error 1/	1, 2, 3, 4	X
15	3.6.12	4.10.13	Residual voltage	1, 2, 3, 4	X
16	3.6.13	4.10.14	Interaxis error 1/	1, 2, 3, 4	X
17	3.6.14	4.10.15	Impedance	1, 2, 3, 4	X
18	3.6.15.1	4.10.16.1	Compensating residual voltage 1/	1, 2, 3, 4	X
19	3.6.15.2	4.10.16.2	Compensating winding phase 1/	1, 2, 3, 4	X
20	3.6.15.3	4.10.16.3	Compensating winding equality 1/	1, 2, 3, 4	X
21	3.6.16	4.10.17	Security of terminals or wire leads	1, 2, 3, 4	
22	3.6.17	4.10.18	Harmonic distortion	1, 2, 3, 4	
23	3.6.18	4.10.19	Shift of resolver zero	1, 2, 3, 4	
24	3.6.8.2	4.10.9.2	Transformation ratio var./primary voltage	1, 2, 3, 4	
25	3.6.9.2	4.10.10.2	Phase shift var./primary voltage	1, 2, 3, 4	
26	3.6.19	4.10.21	Resonant frequency response	1, 2, 3, 4	
27	3.6.20	4.10.22	Temperature rise	1, 2, 3, 4	
28	3.6.21	4.10.23	Electromagnetic interference	1, 2, 3, 4	
29	3.7.1	4.10.24	Vibration, followed by test nos. 1, 13, 14, 15, 4, 3, 5 & 6	1, 2, 3, 4	
30	3.7.2.1	4.10.25.1	Shock, specified pulse, followed by test nos. 1, 13, 14, 15, 4, 3, 5 & 6	1, 2, 3, 4	

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TABLE II - QUALIFICATION AND QUALITY CONFORMANCE INSPECTION (Cont'd)

TEST NO.	REQ'T	TEST	TEST METHOD OR EXAMINATION	INSPECTION	
				QUALIFICATION Sample Nos.	QUALITY CONFORMANCE
31	3.7.3.1	4.10.26.1	Altitude, low temp. during which test nos. 1 and 6 shall be performed	1, 2	
32	3.7.3.2	4.10.26.2	Altitude, high temp. during which test nos. 1, 6 and 27 shall be performed	1, 2	
33	3.7.4	4.10.27	Endurance, following which test nos. 1, 3, 13, 14, 15 and 4 shall be performed	1, 2	
34	3.7.5.1	4.10.28.1	Ambient low temp. during which test nos. 1, 5, 6, 13, 14, 15 and 4 shall be performed	3, 4	
35	3.7.5.2	4.10.28.2	Ambient high temp. during which test nos. 1, 13, 14, 15, 4, 5 and 6 shall be performed	3, 4	
36	3.7.6	4.10.29	Moisture resistance, followed by test nos. 1, 4, 13, 14, 5 and 6	3, 4	
37	3.7.7	4.10.30	Explosion resistance 1/	1, 2	
38	3.7.8	4.10.31	Salt atmosphere 1/	1, 2	
39	3.7.2.2	4.10.25.2	Shock, high impact, followed by test nos. 1, 5, 6, 13, 14, 15, 4 and 3	1, 2, 3, 4	

NOTE: 1/ Tests no. 13 function error, No. 14 electrical error, No. 16 interaxis error, No. 18 compensating residual voltage, No. 19 compensating winding phase, No. 20 compensating winding equality, No. 37 explosion resistance and No. 38 salt atmosphere shall be performed when specified by the applicable specification sheet.

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4.10 Test methods and examinations.

4.10.1 Variation of brush contact resistance.- The variation of brush contact resistance between terminals R2R4 and R1R3 of the resolver shall be tested in accordance with MIL-S-81963 and shall meet the requirements of 3.6.1.

4.10.2 Visual and mechanical examination.- The resolver shall be examined in accordance with MIL-S-81963 and shall meet the requirements therein.

4.10.3 Shaft radial and end play.- Shaft radial and end play shall be tested in accordance with MIL-S-81963 and shall meet the requirements of 3.6.2.

4.10.4 Mechanical breakaway torque.- Mechanical breakaway torque shall be tested in accordance with MIL-S-81963 and shall meet the requirements of 3.6.3.

4.10.5 Dielectric withstanding voltage.- Dielectric withstanding voltage shall be tested in accordance with MIL-S-81963 and Table III herein and shall meet the requirements of 3.6.4.

TABLE III - Dielectric Withstanding Voltages and Application Points

Maximum Rated Voltages, rms	Initial Acceptance Test Voltages, rms (50 or 60 Hz)		Subsequent Test Voltages, rms (50 or 60 Hz)	
	Each Winding to Housing & Primary to Secondary Windings	Between Electrically Separate Windings in Intimate Contact	Each Winding to Housing & Primary to Secondary Windings	Between Electrically Separate Windings in Intimate Contact
Up to 50	242 to 250	242 to 250	194 to 200	194 to 200
51 to 100	485 to 500	242 to 250	388 to 400	194 to 200
101 to 200	870 to 900	242 to 250	720 to 740	194 to 200

4.10.6 Insulation resistance.- Insulation resistance shall be tested in accordance with MIL-S-81963 and Table III herein and shall meet the requirements of 3.6.5.

4.10.7 Current.- The current shall be tested in accordance with MIL-S-81963. The resolver shall be brought to the stabilized operating temperature condition of 4.5 and the current drawn by each primary winding when energized in accordance with 4.6 and with all other windings open-circuited, shall meet the requirements of 3.6.6.

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4.10.8 Power.- The power shall be measured in accordance with MIL-S-81963. The resolver shall be brought to the stabilized operating temperature condition of 4.5 and the power consumed by each primary winding when energized in accordance with 4.6 and with all other windings open-circuited, shall meet the requirements of 3.6.7.

4.10.9 Transformation ratio.

4.10.9.1 Transformation ratio.- The resolver shall be mounted in accordance with 4.3.1, brought to the stabilized operating temperature condition of 4.5, energized in accordance with 4.6 and set at resolver zero in accordance with the applicable figure of 3.4.3. The resolver shall then be connected and the rotor positioned in accordance with Table IV as applicable. The secondary winding voltage at the position of maximum coupling shall be measured with a fundamental frequency rms voltage indicating instrument that shall not alter the open-circuited secondary voltage by more than 0.05 percent. The ratio of this secondary winding voltage to the primary energization winding voltage is the transformation ratio which shall be in accordance with the requirements of 3.6.8.1.

4.10.9.2 Transformation ratio variation with voltage.- The transformation ratio shall be measured at the maximum and minimum operating voltages specified in the applicable specification sheet under the test conditions of 4.10.9.1 and shall meet the requirements of 3.6.8.2.

4.10.9.3 Equality of transformation ratios.- The maximum difference between measured transformation ratios of 4.10.9.1 shall be in accordance with the requirements of 3.6.8.3.

4.10.10 Phase shift.

4.10.10.1 Phase shift.- The resolver shall be mounted in accordance with 4.3.1, brought to the stabilized operating temperature condition of 4.5, energized in accordance with 4.6 and set at resolver zero in accordance with the applicable figure of 3.4.3. The resolver shall then be connected and the rotor positioned in accordance with Table IV as applicable. The phase shift shall be measured at the position of maximum coupling to an accuracy of $\pm$ 0.10 degrees with an instrument having a primary impedance of not less than 500K ohms resistance shunted by a 30pF capacitance. The phase shift shall be in accordance with the requirements of 3.6.9.2.

4.10.10.2 Phase shift variation with voltage.- The test of 4.10.10.1 shall be repeated using the minimum and maximum operating voltages specified in the applicable specification sheet. The phase shift variation with primary voltage change shall be in accordance with the requirements of 3.6.9.2.

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TABLE IV. TR EQUALITY WINDING COMBINATIONS

Primary Energization Winding	Shorted Winding	Secondary Winding	Rotor Angle
S1S3	S2S4, C2C4	R1R3	0°
S1S3	S2S4, C2C4	R2R4	270°
S2S4	S1S3, C1C3	R2R4	0°
S2S4	S1S3, C1C3	R1R3	90°
R1R3	R2R4	S1S3	0°
R1R3	R2R4	S2S4	90°
R2R4	R1R3	S2S4	0°
R2R4	R1R3	S1S3	270°

TR EQUALITY-ROTOR/COMP. WINDINGS BY CALCULATION

R/C Winding Calculation Based on R/S and C/S Measured Values	Rotor Angle
$\frac{\frac{R1R3}{S1S3}}{\frac{C1C3}{S1S3}} = \frac{R1R3}{C1C3}$	0°
$\frac{\frac{R2R4}{S2S4}}{\frac{C2C4}{S2S4}} = \frac{R2R4}{C2C4}$	0°
$\frac{\frac{R1R3}{S2S4}}{\frac{C2C4}{S2S4}} = \frac{R1R3}{C2C4}$	90°
$\frac{\frac{R4R2}{S1S3}}{\frac{C1C3}{S1S3}} = \frac{R4R2}{C1C3}$	90°

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4.10.11 Function error.

4.10.11.1 Stator energized.- The resolver shall be mounted in accordance with 4.3.1, brought to the stabilized operating temperature condition of 4.5 and terminals S1S3 shall be energized in accordance with 4.6. The rotor shall first be set to the approximate zero position by aligning the shaft and housing zero indexes, and then to the exact zero position by obtaining a null across terminals R2R4 using the circuit of Figure 2. The resolver shall be connected as shown in Figure 3 and the rotor angle set at 90 degrees with the "read-out" transformer set to 1.00000. The normalizing transformer and phase shift capacitor shall be adjusted to give the minimum null voltage reading of the phase sensitive nullmeter. The normalizing transformer and phase shift capacitor shall not be readjusted throughout the remainder of the test. The "read-out" transformer setting that results in a minimum reading of the phase sensitive nullmeter minus the sine of the rotor angle times 100 gives the function error as a percentage of maximum output. The function error shall be determined at 5 degree rotor angle increments from zero through 180 degrees and the test shall be repeated substituting the following terminal connections: S2S4 for S1S3; R1R3 for R2R4. The function error shall be in accordance with the requirements of 3.6.10.

4.10.11.2 Rotor energized.- The test conditions of 4.10.11.1 shall apply to rotor energized resolvers except for the following terminal substitutions: R1R3 for S1S3; R2R4 for S2S4; S1S3 for R1R3; S2S4 for R2R4. The function error shall be in accordance with the requirements of 3.6.10.

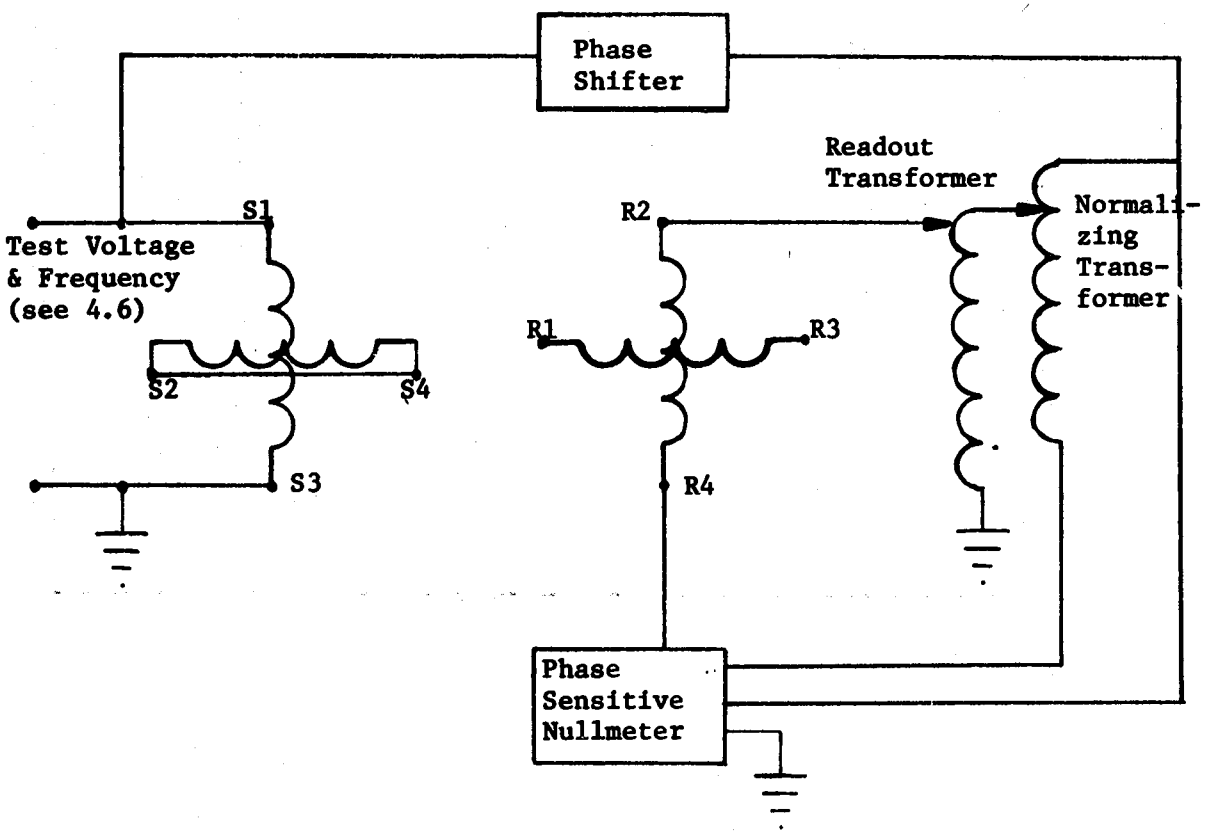


Figure 3. Function Error Test Circuit.

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4.10.12 Electrical error.- Resolvers shall be tested for electrical error by either of the proportional voltage methods described in 4.10.12.1 or 4.10.12.2. The electrical error shall be measured and recorded at rotor angular increments of not more than 5 degrees from zero through 360 degrees, and shall be in accordance with the requirements of 3.6.11.

4.10.12.1 Proportional voltage nulling method.

4.10.12.1.1 Stator error, rotor energized resolvers.- The resolver shall be mounted in accordance with 4.3.1, brought to the stabilized operating temperature condition of 4.5 and terminals R1R3 shall be energized in accordance with 4.6. The rotor shall first be set to the approximate zero position using the index marks on the shaft and housing, and then to the exact zero position using the circuit of Figure 1. The resolver shall then be connected to the appropriate test circuit of Figure 4 or 5 and the error shall be determined.

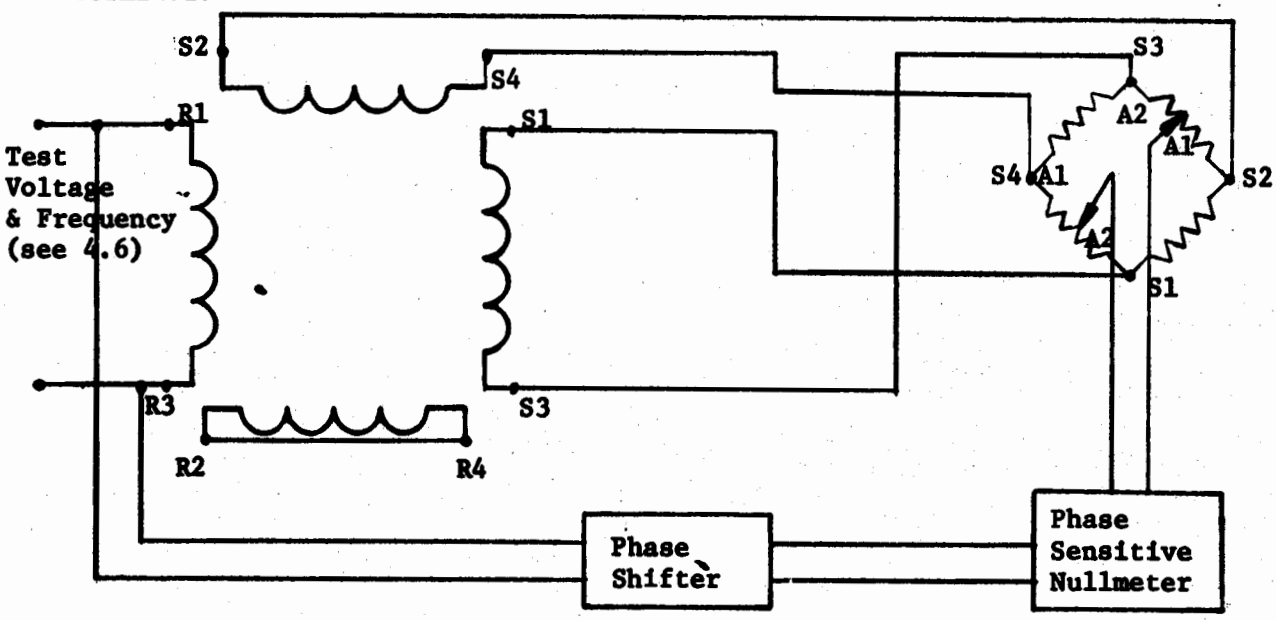


Figure 4. Basic Circuit for Stator Electrical Error Test, Rotor Energized Resolvers, Proportional Voltage Null Method.

The electrical error of the resolver is the deviation from the (mechanical) rotor angle to balance the bridge for the following ratios:

$\frac{A1}{A2} = \tan\theta$, for $\theta = \frac{n\pi}{2} - \pi/4$ to $\frac{n\pi}{2} + \pi/4$, for n positive even integers (n = 0, 2, 4, 6, etc.)

$\frac{A2}{A1} = \cot\theta$, for $\theta = \frac{n\pi}{2} - \pi/4$ to $\frac{n\pi}{2} + \pi/4$, for n positive odd integers (n = 1, 3, 5, 7, etc.)

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Tan	Cot	A1/A2, A2/A1
0°	90°	0.000000
5	85	0.087488
10	80	0.176327
15	75	0.267949
20	70	0.363970
25	65	0.466307
30	60	0.577350
35	55	0.700207
40	50	0.839099
45	45	1.000000

In the bridge of Figure 4 the resistor arms S3S4, S1S2 and the divider arms S3S2 and S1S4 shall be non-inductive resistors of nominal value 10K ohms and shall be equal to within 1 ohm. The angle taps of the dividers A1/A2, A2/A1 shall be within 0.005 percent of the theoretical values stated. The phase shift of the divider shall be less than 10 minutes of arc at the test frequency. The phase sensitive nullmeter shall have an input impedance not less than 500K ohms shunted by 30pF. The minimum indication of the phase sensitive nullmeter shall be less than the voltage equivalent to 0.2 minute of arc displacement from the null position of the resolver under test when energized in accordance with 4.6. The phase sensitive nullmeter rejection shall be such that a quadrature input of 0.2 percent of the test resolver maximum voltage output, coupled with an harmonic input of 1 percent of the test resolver maximum voltage input, shall not produce a greater meter indication than a test resolver rotor displacement of 0.2 minutes of arc from the null position.

4.10.12.1.2 Stator error, stator energized resolvers. The basic circuit for electrical testing of stator error of stator energized resolvers is shown in Figure 5. The stator terminals of the stator energized resolver under test are connected to the stator terminals of the rotor energized resolver, of equal or larger frame size, which has successfully completed the electrical error tests of 4.10.12.1 and which is energized in accordance with 4.6. The bridge voltage dividers are to be set to the ratios appropriate to the rotor angle of the resolver under test (defined in 4.10.12.1.1) and the energizing resolver shaft position adjusted to null the bridge output. The shaft position of the resolver under test is adjusted to null its phase sensitive nullmeter (No. 2). The electrical error is the deviation of the rotor angle from the value used to set the bridge voltage divider ratios.

4.10.12.1.3 Rotor error, stator energized resolvers. The test 4.10.12.1.1 for stator error of rotor energized resolvers shall apply to rotor error of stator energized resolvers with the following terminal substitutions: S1S3 for R1R3; S2S4 for R2R4; R1R3 for S1S3; and R2R4 for S2S4.

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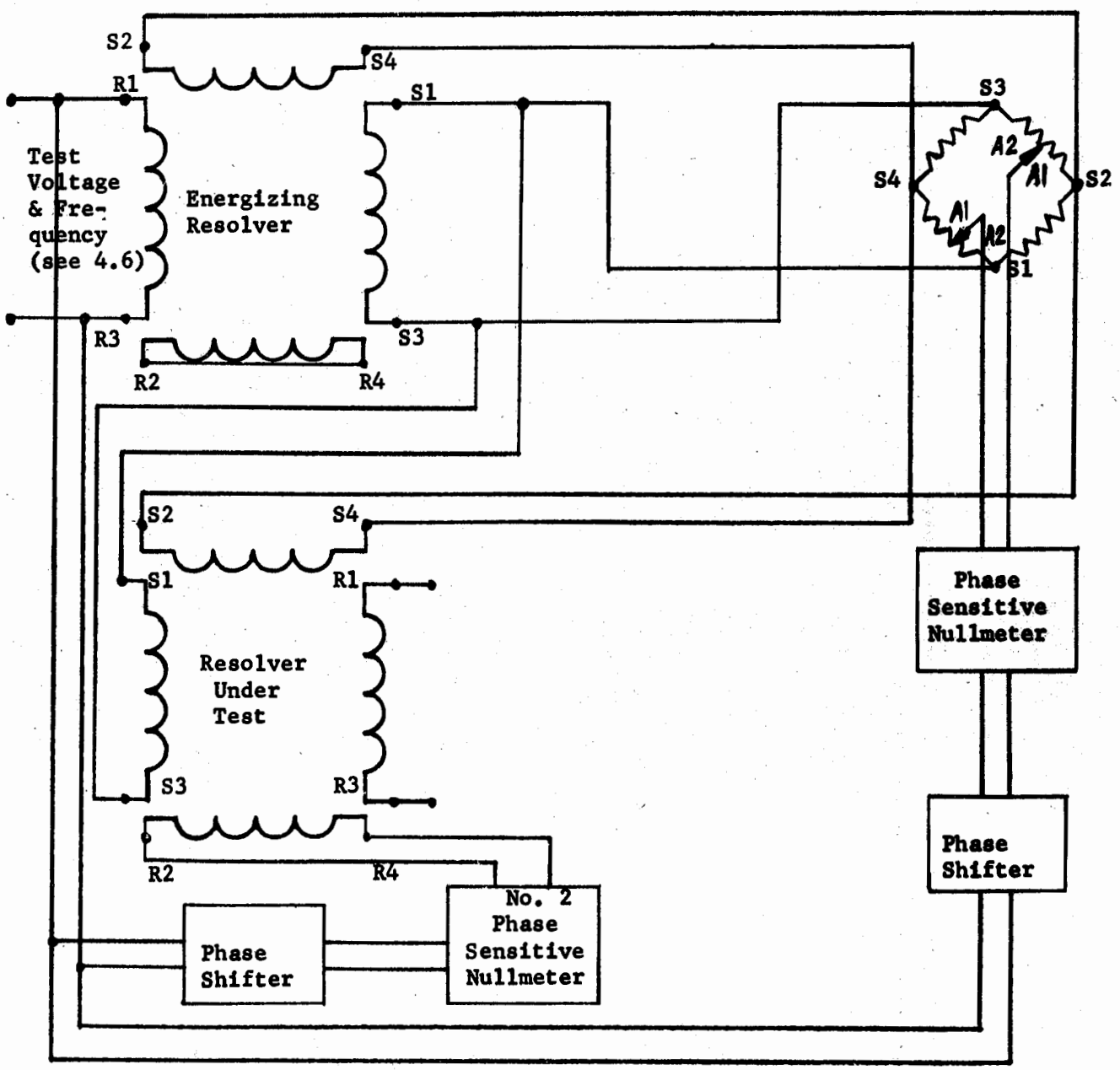


Figure 5. Basic Circuit for Stator Electrical Error Test, Stator Energized Resolvers, Proportional Voltage Null Method.

4.10.12.2 Proportional voltage gradient method.

4.10.12.2.1 Stator error, rotor energized resolvers.- The basic circuit is shown in Figure 6. The circuit and operation are similar to that of 4.10.12.1.1 excepting that the phase sensitive null meter has been replaced by a calibrated phase sensitive amplifier with compensating network, an indicator and recorder. The calibrated phase sensitive amplifier with compensating network shall meet the loading and quadrature voltage rejection requirements (for the phase sensitive null meter) of 4.10.12.1.1

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and shall have constant voltage gradient (defined as electrical output per unit rotor deviation) at all bridge settings. The resolution of the indicator and recorder shall be such that an electrical error of 30 arc seconds may be determined without interpolation. The electrical error for a given rotor position is defined as the indicator/recorder reading divided by the voltage gradient when the bridge voltage dividers are set to the appropriate ratios for that rotor angle as defined in 4.10.12.1.1.

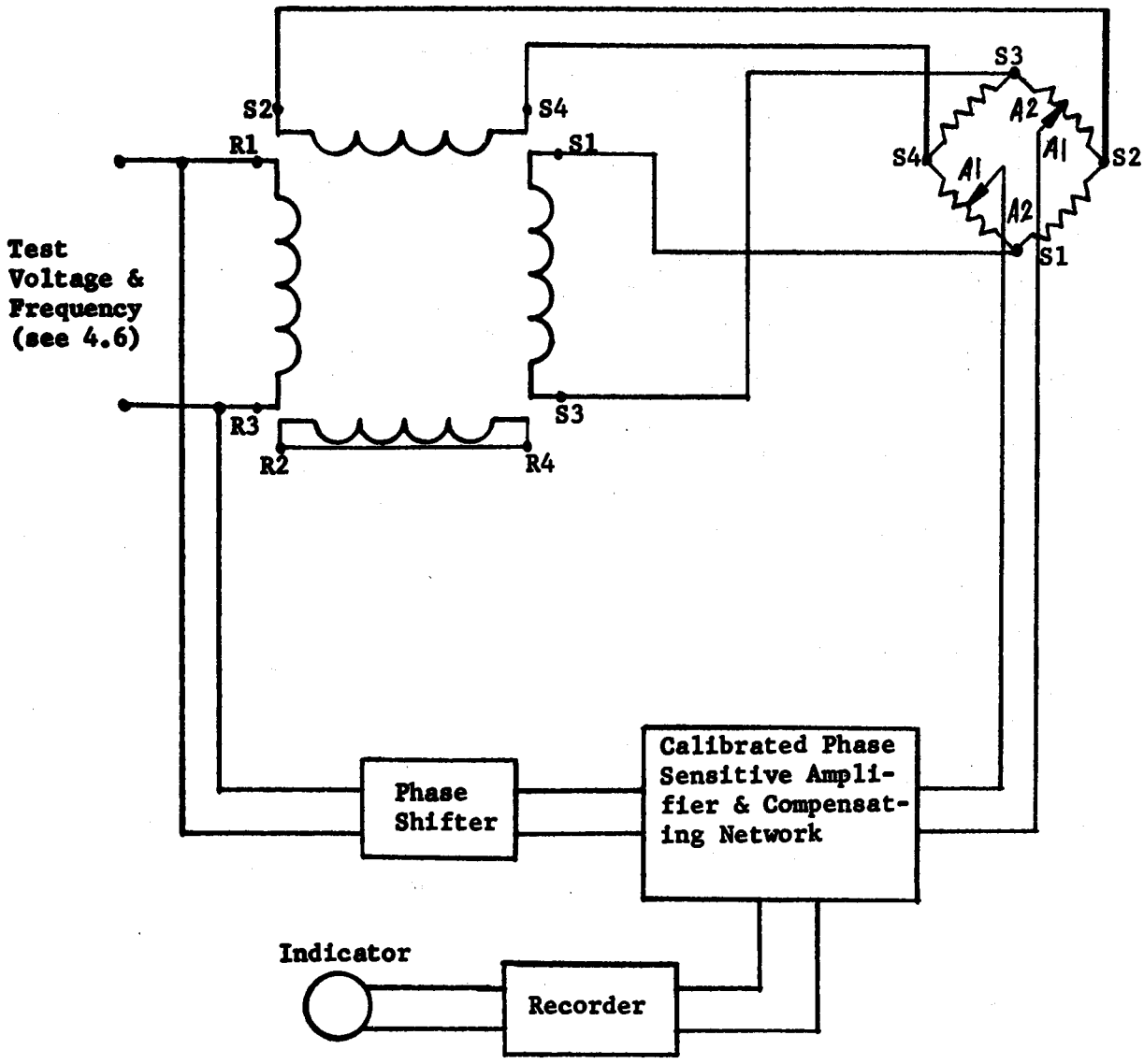


Figure 6. Basic Circuit for Stator Electrical Error Test, Rotor Energized Resolvers, Proportional Voltage Gradient Method.

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4.10.12.2.2 Stator error, stator energized resolvers.- The basic circuit is shown in Figure 7. The circuit and operation are similar to that of 4.10.12.1.2 excepting that phase sensitive null meter No. 2 has been replaced by a phase sensitive calibrated amplifier with compensating network indication and recorder as defined in 4.10.12.2.1. The resolver under test is set to the rotor position at which the electrical error is to be measured and the bridge and energizing resolver set up and adjusted appropirately for the rotor position of the resolver under test as described in 4.10.12.1.2. The electrical error is the indicator/recorder reading divided by the voltage gradient as defined in 4.10.12.2.1.

4.10.12.2.3 Rotor error, stator energized resolvers.- The test shall be carried out as in 4.10.12.2.1 and Figure 6 (stator error, rotor energized resolvers) with the following terminal substitutions: S1S3 for R1R3; S2S4 for R2R4; R1R3 for S1S3; and R2R4 for S2S4.

4.10.13 Residual voltage.- Fundamental and total residual voltages shall be measured across the terminals and at the angles specified in Table V. The resolver shall be mounted in accordance with 4.3.1, brought to the stabilized operating temperature condition of 4.5 and energized in accordance with 4.6. These residual voltages may be measured by method 4.10.13.1 or 4.10.13.2. The voltage measuring instruments used shall indicate the value in terms of the rms value of an equivalent sine wave and shall have an input impedance not less than 500K ohms shunted by 30pF. The fundamental and total residual voltages shall be in accordance with the requirements of 3.6.12. For qualification the test shall also be performed at the maximum operating voltage specified by the applicable specification sheet.

4.10.13.1 Frequency sensitive voltmeter method.- The frequency sensitive voltmeter to be used shall have a filter characteristic such that the output changes by not greater than $\pm$ 0.5 percent for a 1 percent change in frequency from the center frequency of the filter and by -97 percent (-30 dB) for a one octave change from the center frequency of the filter. The resolver shall be calibrated to allow for the center frequency insertion loss of the filter. The resolver rotor shall be turned to give a minimum reading on the frequency sensitive voltmeter. This is the fundamental component of the residual voltage. The voltage read by a non-frequency-sensitive voltmeter at the same (undisturbed) rotor position is the total residual voltage.

4.10.13.2 Phase sensitive voltmeter method.- The resolver rotor shall be turned until the in-phase component of the residual voltage, as indicated by the phase sensitive voltmeter, is zero. The quadrature reading is the fundamental component of the residual voltage. The voltage read by a non-phase-sensitive voltmeter is the total residual voltage.

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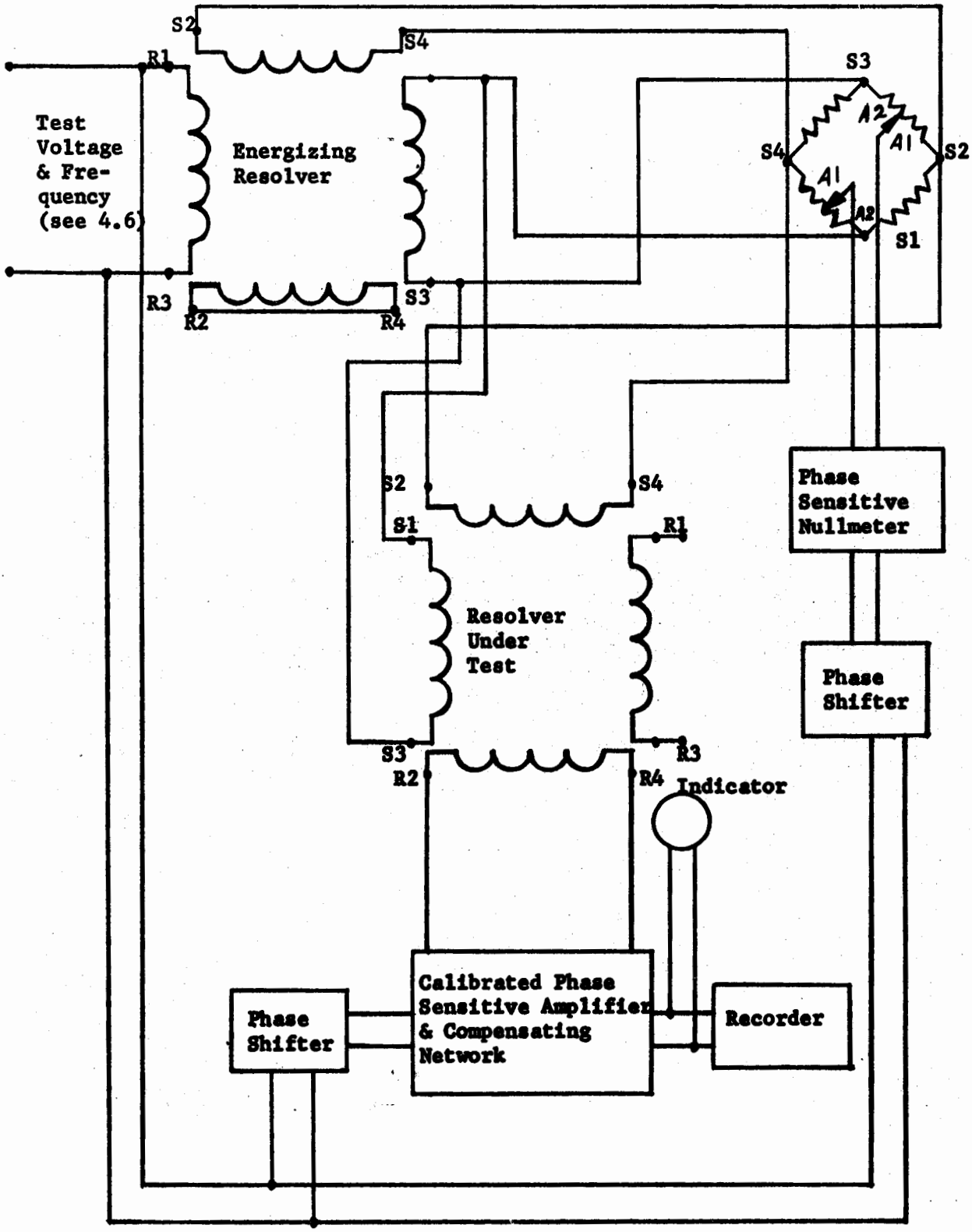


Figure 7. Basic Circuit for Stator Electrical Error Test, Stator Energized Resolvers, Proportional Voltage Gradient Method.

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TABLE V

Residual Voltage and Interaxis Error. Connections and Angular Displacements				
Unit	Primary Voltage	Measurement Points	Rotor Electrical Angle	Interaxis Serial
Rotor Energized	R1R3 (R2R4 Shorted)	S2S4	0°	A
		S2S4	180°	B
		S1S3	90°	C
		S1S3	270°	D
	R2R4 (R1R3 Shorted)	S1S3	0°	E
		S1S3	180°	F
		S2S4	90°	G
		S2S4	270°	H
Stator Energized	S1S3 (S2S4 Shorted)	R2R4	0°	A
		R2R4	180°	B
		R1R3	90°	C
		R1R3	270°	D
	S2S4 (S1S3 Shorted)	R1R3	0°	E
		R1R3	180°	F
		R2R4	90°	G
		R2R4	270°	H

4.10.14 Interaxis error.- The resolver shall be mounted in accordance with 4.3.1, brought to the stabilized operating temperature condition of 4.5 and with each of the appropriate primary windings energized in turn in accordance with 4.6, the deviation of the null position from the nominal rotor electrical angles specified in Table V (Interaxis Serial A through H) shall be recorded. The unenergized primary winding shall be shorted. The following interaxis errors (algebraic differences between the deviations) shall be tabulated:

Secondary winding; A-C, B-D, E-G, F-H

Primary winding; A-G, B-H, C-F, D-E

Resolver (overall); A-E, B-F, C-G, D-H

The interaxis errors shall be in accordance with the requirements of 3.6.13.

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4.10.15 Impedance.- Impedance shall be measured in accordance with MIL-S-18963 and Table VI herein. The resolver shall be brought to the stabilized operating temperature of 4.5 and energized in accordance with 4.6 and then set to resolver zero. The impedance shall then be measured and shall be in accordance with the requirements of 3.6.14.

TABLE VI

Impedance Measurements			
Impedance	Terminals for Impedance Measurements (each winding tested separately)	Voltage across Resolver Terminals	Additional Connections
Zro	R1R3, R2R4	see 4.6	1/
Zso	S2S4, S1S3	see 4.6	1/
Zrs	R1R3, R2R4	2/	3/
Zss	S2S4, S1S3	4/	3/
Zco	C2C4, C1C3	see 4.6	5/

NOTES: 1/ Remaining energizing winding short circuited and all other windings open circuited.

2/ The voltage necessary to induce a current within ± 3% of current in Zro measurement.

3/ All other windings except compensating windings short circuited.

4/ The voltage necessary to induce a current within ± 3% of current in Zso measurement.

5/ Remaining compensating winding short circuited and all other windings open circuited.

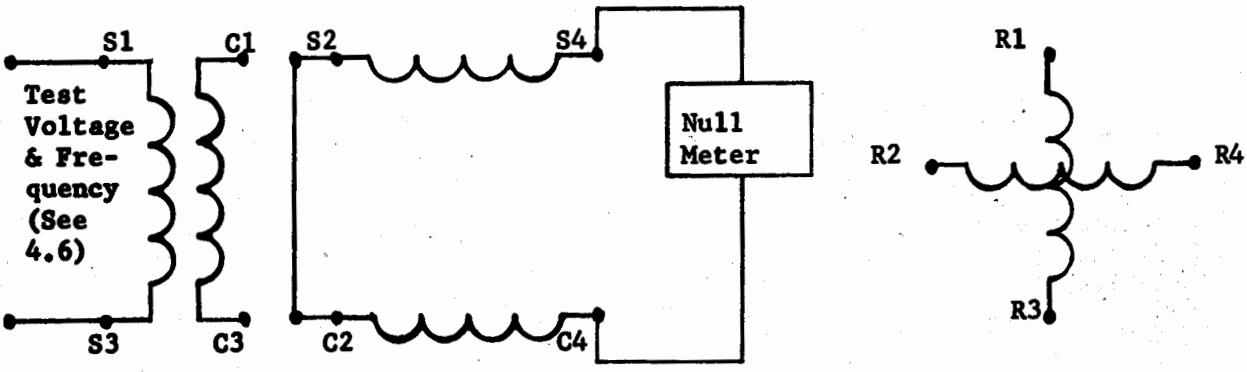
4.10.16 Compensating windings.

4.10.16.1 Residual voltage.- When the resolver is connected as shown in Figure 8, the reading of the null meter shall be in accordance with the requirements of 3.6.15.1.

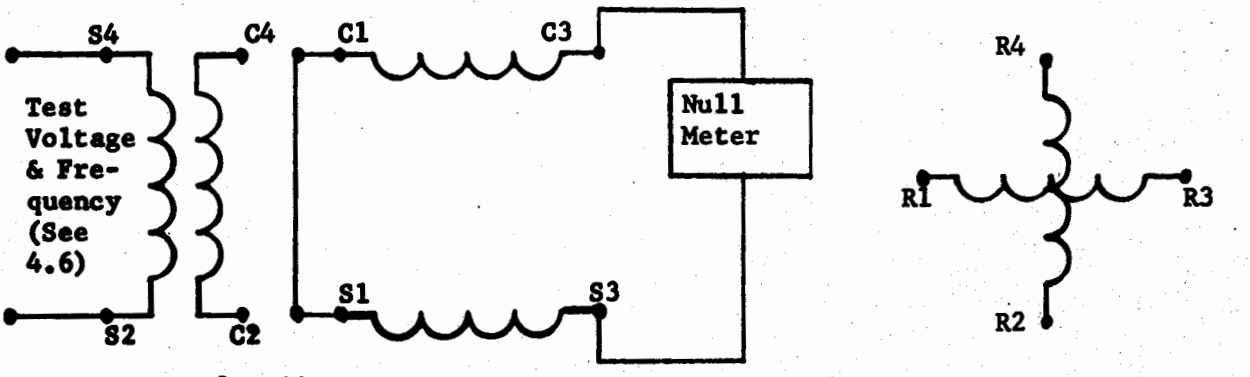
4.10.16.2 Winding phase.- When the resolver is connected as shown in Figure 9, the reading of the voltmeter shall be less than the primary test voltage. Winding phase shall be in accordance with the requirements of 3.6.15.2.

4.10.16.3 Equality of compensating windings.- When the resolver is connected as shown in Figure 10 the millivolt values obtained shall be in accordance with the requirements of 3.6.15.3.

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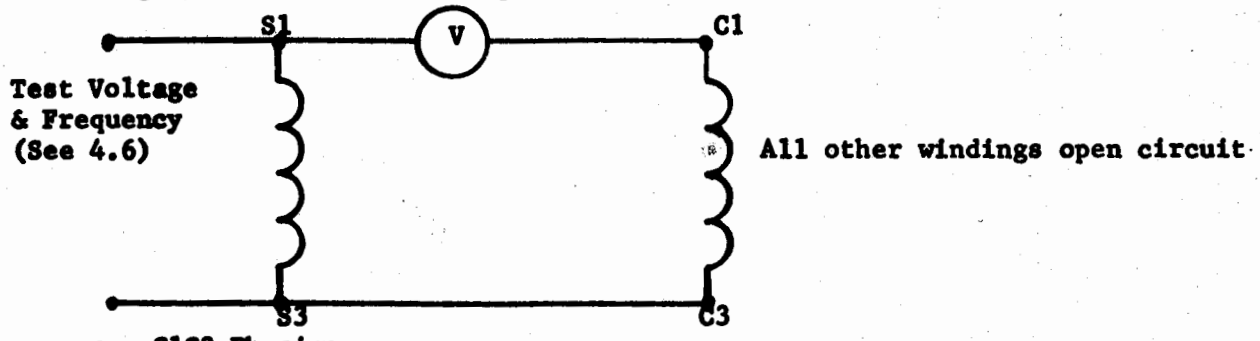


a. C2C4 Null

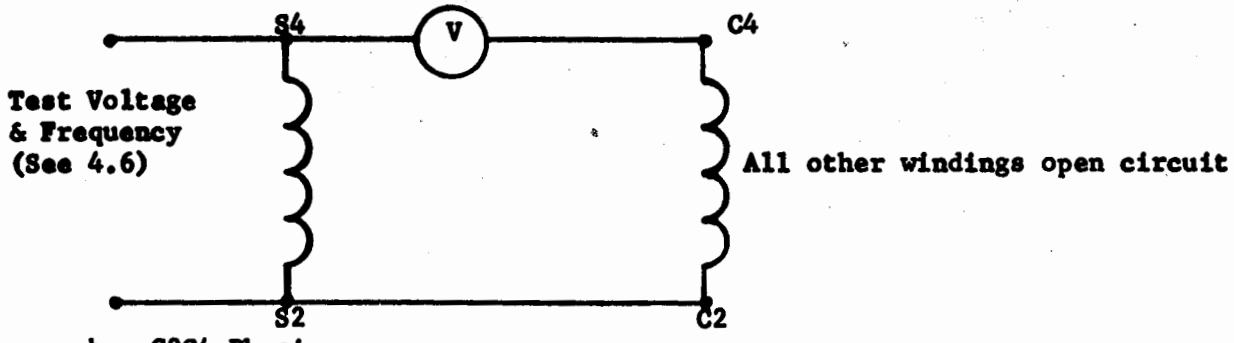


b. C1C3 Null

Figure 8. Residual Voltage Test Circuit, Compensating Windings.



a. C1C3 Phasing



b. C2C4 Phasing

Figure 9. Winding Phasing Test Circuit, Compensating Windings.

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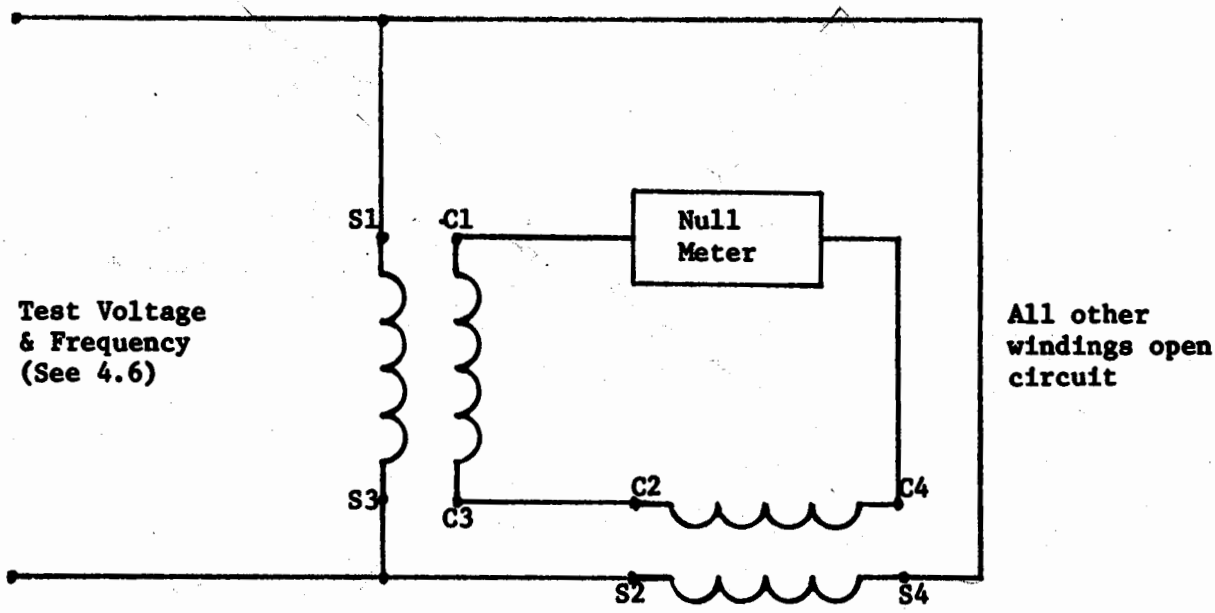


Figure 10. Equality of Compensating Windings Test Circuit.

4.10.17 Security of terminals or wire leads.- The security of each screw type or solder pin-type of terminal or of each wire lead, shall be tested in accordance with MIL-S-81963 and shall meet the requirements of 3.6.16.

4.10.18 Harmonic distortion.- The resolver shall be mounted in accordance with 4.3.1, brought to the stabilized operating temperature condition of 4.5 while energized in accordance with 4.6 and set at exact resolver zero as specified in 3.4.3.

4.10.18.1 Rotor energized resolvers.- After compliance with 4.10.18, the resolver shall then be connected as shown in Figure 11 and energized at the maximum operating voltage specified in the applicable specification sheet. The total harmonic distortion is the rms value of the difference of the amplitudes up to and including the 9th harmonic (expressed as a percentage of the energization voltage) in the energization voltage supply waveform and the S1S3 output, and shall be as specified in 3.6.17.

4.10.18.2 Stator energized resolvers.- Harmonic distortion shall be determined as in 4.10.18.1 except for the following terminal substitutions: S1S3, S2S4 for R1R3, R2R4; R1R3, R2R4 for S1S3, S2S4.

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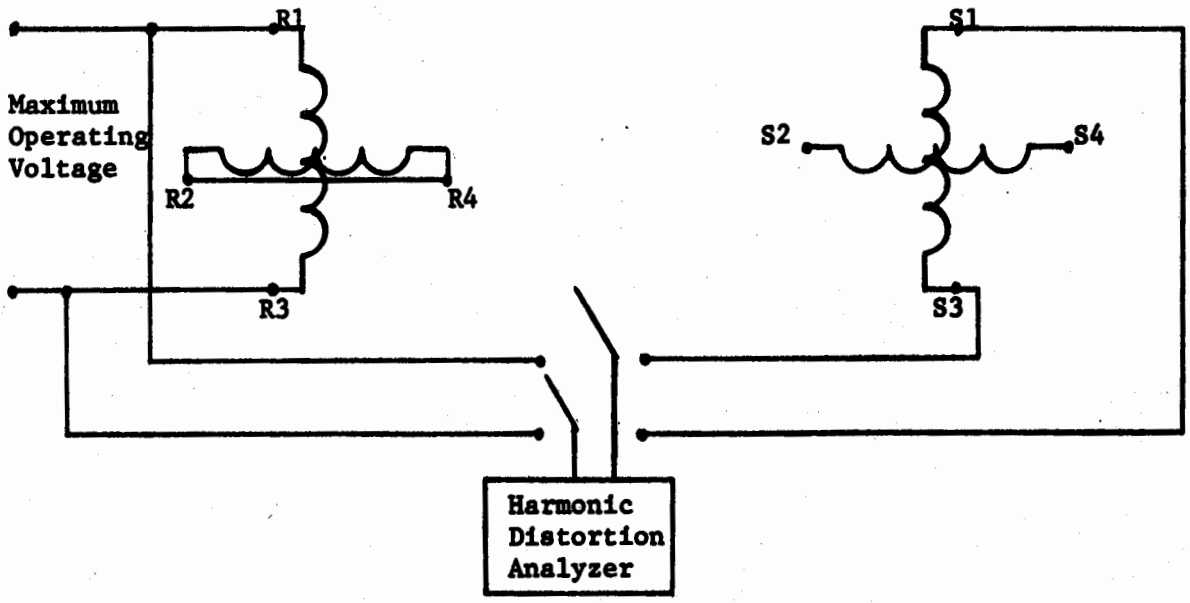


Figure 11. Harmonic Distortion Test Circuit.

4.10.19 Shift of resolver zero.

4.10.19.1 Variation with voltage.- The resolver shall be mounted in accordance with 4.3.1, brought to the stabilized operating temperature condition of 4.5 while energized in accordance with 4.6, and set at the exact zero, as specified in 3.4.3. Energization shall then in turn be changed to the maximum and minimum primary voltages and the resolver zeroed at each of these . The shift of resolver zero shall be in accordance with the requirements of 3.6.18.2.

4.10.19.2 Variation with frequency.- The test of 4.10.19.1 shall be repeated with the test voltage held constant in accordance with 4.6 and the resolver reset at zero when the operating frequency is first decreased by 10 percent and then increased by 10 percent. The shift in resolver zero shall be in accordance with the requirements of 3.6.18.2.

4.10.20 Resolver zero marking.

4.10.20.1 Rotor energized resolvers.- The resolver windings shall be connected as shown in Figure 12 and the rotor turned until the voltmeter minimum reading is obtained. The relative position of the resolver zero index marked on the housing and the index marked on the shaft shall be in accordance with the requirements of 3.5.2.

4.10.20.2 Stator energized resolvers.- Resolver zero shall be determined as in 4.10.20.1 except for the following terminal substitutions: S1S3, S2S4 for R1R3, R2R4; and R1R3, R2R4 for S1S3, S2S4.

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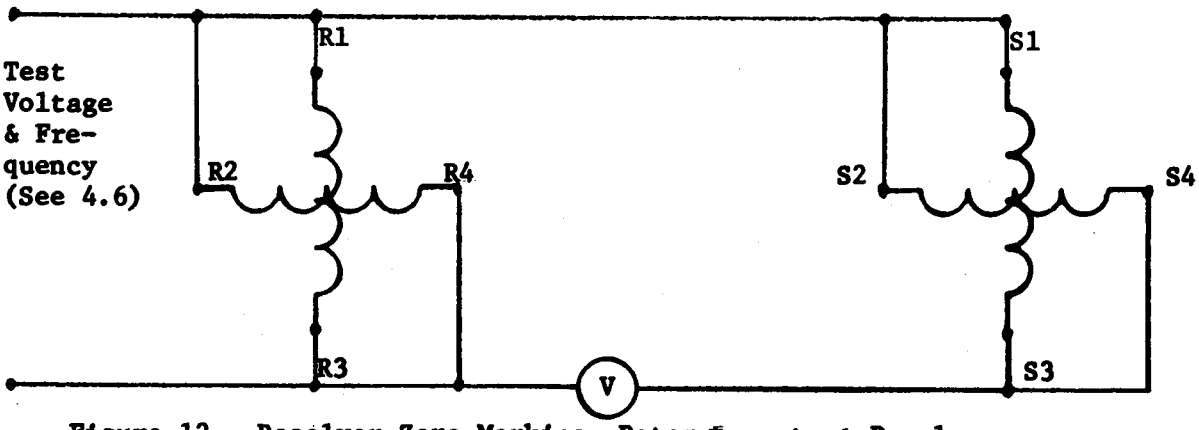


Figure 12. Resolver Zero Marking, Rotor Energized Resolver

4.10.21 Resonant frequency response.- The resolver shall be mounted in accordance with 4.3.1, brought to the stabilized operating temperature condition of 4.5 with one primary winding energized in accordance with 4.6. The rotor shall be set at the position of maximum coupling nearest resolver zero. The applicable secondary winding shall be monitored to ascertain the resolver has attained maximum voltage while simultaneously experiencing a -90 degree phase shift. The resonant frequency response shall be identified and recorded. The same procedure shall be repeated with the exception that the other primary winding shall be energized and tested separately. The resolver shall meet the minimum requirement for resonant frequency response for each primary winding combination in accordance with the requirements of 3.6.19.

4.10.22 Temperature rise.- The resolvers shall be tested in accordance with MIL-S-81963. After the resolver has attained the stabilized non-operating temperature condition of 4.4, the dc resistance of one of the secondary windings shall be measured. Both primary windings shall then be energized in accordance with 4.6, with the secondary windings unloaded. The dc resistance of the secondary winding measured prior to energization shall again be measured when the resolver has attained the stabilized operating temperature condition of 4.5. The temperature rise shall be in accordance with the requirements of 3.6.20.

4.10.23 Electromagnetic interference.- The resolver shall be subjected to the electromagnetic interference test of MIL-S-81963. Both resolver primary windings shall be energized in accordance with 4.6, and the rotor shall be rotated at 600 $\pm$ 50 rpm. [A load equivalent to four times the secondary winding open circuit impedance shall be applied across both secondary windings.] Compensating windings shall be unloaded. Electromagnetic conducted and radiated interference of the resolver shall be in accordance with the requirements of 3.6.21.

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4.10.24 Vibration.- The resolvers shall be tested in accordance with the requirements of MIL-S-81963. During the test, both rotor windings shall be energized in accordance with 4.6 with their rotor shafts free to rotate while mechanically loaded. In the case of a stator energized resolver, the energizing voltage of both rotor windings shall be determined accordingly by the transformation ratio as specified in 3.6.8. Immediately after the test, each resolver shall be examined for loose or damaged parts and shall then be subjected to and shall meet the tests specified in Table II.

4.10.25 Shock.

4.10.25.1 Shock, specified pulse.- The resolvers shall be tested in accordance with the requirements of MIL-S-81963. During the test, both rotor windings shall be energized in accordance with 4.6 with their rotor shafts free to rotate while mechanically loaded. In the case of a stator energized resolver, the energizing voltage of both rotor windings shall be determined accordingly by the transformation ratio as specified in 3.6.8. Immediately after the test, each resolver shall be examined for loose or damaged parts and shall then be subjected to and shall meet the tests specified in Table II.

4.10.25.2 Shock, high impact.- The resolvers shall be tested in accordance with the requirements of MIL-S-81963. During the test, both rotor windings shall be energized in accordance with 4.6 with their rotor shafts free to rotate while mechanically loaded. In the case of a stator energized resolver, the energizing voltage of both rotor windings shall be determined accordingly by the transformation ratio as specified in 3.6.8. Immediately after the test, each resolver shall be examined for loose or damaged parts and shall then be subjected to and shall meet the tests specified in Table II.

4.10.26 Altitude.

4.10.26.1 Altitude low temperature.- The resolvers shall be tested in accordance with the requirements of MIL-S-81963. After the attainment successively of the stabilized non-operating and stabilized operating temperature conditions as defined in 4.4 and 4.5 and the specified reduction of chamber pressure, the resolvers shall be energized in accordance with 4.6. While still in the specified environment, the resolvers shall be subjected to and shall meet the tests specified in Table II.

4.10.26.2 Altitude high temperature.- The resolvers shall be tested in accordance with the requirements of MIL-S-81963. The internal ambient temperature of the test chamber shall be raised to and controlled at 125 $\pm$ 2°C. After the attainment successively of the stabilized non-operating and stabilized operating temperature conditions as defined in 4.4 and 4.5 and the specified reduction of chamber pressure, the resolver shall be energized in accordance with 4.6. While still in the specified environment, the resolvers shall be subjected to and shall meet the tests specified in Table II.

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4.10.27 Endurance.- The resolvers shall be tested for 2000 hours in accordance with the requirements of MIL-S-81963 while energized in accordance with 4.6 and while rotating the rotor at 1150 ± 50 rpm. After completion of the endurance test, the resolvers shall be subjected to and shall meet the tests specified in Table II.

4.10.28 Ambient temperature.

4.10.28.1 Ambient low temperature.- The resolvers shall be tested in accordance with the requirements of MIL-S-81963. After having been allowed to attain the stabilized non-operating temperature condition as defined in 4.5, and at the end of the one hour period at -55 ± 2°C, the resolvers shall be allowed to attain the stabilized operating temperature condition as defined in 4.5 and while still in this low temperature environment the resolvers shall be subjected to and shall meet the tests specified in Table II.

4.10.28.2 Ambient high temperature.- The resolvers shall be tested in accordance with the requirements of MIL-S-81963. The resolvers shall be allowed to attain the stabilized non-operating temperature condition as defined in 4.4 at 125 ± 2°C ambient temperature. The resolvers shall then be allowed to attain the stabilized operating temperature conditions as defined in 4.5 at 125 ± 2°C ambient temperature and while still in this environment shall be subjected to and shall meet the tests specified in Table II.

4.10.29 Moisture resistance.- The resolvers shall be tested in accordance with the requirements of MIL-S-81963. With each resolver shaft horizontal, one resolver of the sample shall be energized in accordance with 4.6 and the other resolver shall be unenergized. After completion of the final 24 hour recovery period, the resolvers shall be subjected to and shall meet the tests specified in Table II.

4.10.30 Explosion resistance.- If specified by the applicable specification sheet, the resolvers shall be tested in accordance with the requirements of MIL-S-81963.

4.10.31 Salt atmosphere.- If specified by the applicable specification sheet, the resolver shall be tested in accordance with the requirements of MIL-S-81963.

4.10.32 Inspection of preparation for delivery.- Inspection of preparation for delivery shall be in accordance with MIL-S-81963.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-S-81963. (See 6.2).

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6. NOTES

6.1 Intended use.- The resolvers are intended for use in applications which require performance of trigonometric computations, conversion between rectangular and polar coordinates, and transmission and reception of angular position data by one or more rotor shafts.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Title, number, and date of MIL-S-81963 (see 1.1).
c. Title, number, and date of the applicable specification sheet, the complete nomenclature (see 1.2.1 and 3.1).
d. Whether first article inspection is required (see 3.3).
e. Levels of packaging and packing required (see 5.1).

6.3 First article inspection.- Information pertaining to first article inspection of products covered by this specification should be obtained from the procuring activity for the specific contracts involved.

6.4 Definitions.- For the purpose of this specification, the following definitions apply:

6.4.1 Electrical degree.- An electrical degree is an arbitrary unit of time required for an alternating current to complete 1/360 of one cycle of the fundamental component of the primary voltage.

6.4.2 Electrical error.- The electrical error is the difference between the rotor angle and the electrical angle at any rotor position. The tangent of the electrical angle is calculated from the secondary voltage of both secondary windings when one primary winding is energized. Electrical error is measured in minutes of arc.

6.4.3 Function error.- Function error is the difference (expressed as a percentage of the maximum in-phase component of the secondary voltage) between the in-phase component of the voltage of one secondary winding with one primary winding energized and the theoretical value of the secondary voltage. It is equivalent to

$$\left(\frac{E\theta}{E(\theta = 90^{o})} - \mathrm{Sin}\theta\right) X100$$

Where: $E\theta$ = the measured rms value of the fundamental component of secondary voltage at rotor angle θ.
$E(\theta = 90^{o})$ = the measured rms value of the fundamental component of secondary voltage at $\theta = 90^{o}$.

6.4.4 Harmonic distortion.- Harmonic distortion is the Fourier difference between the applied primary voltage waveform and the open circuit secondary voltage waveform.

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6.4.5 Impedance.

6.4.5.1 Rotor impedance Zro.- The rotor impedance Zro is the impedance of the rotor with the stator terminals open circuited.

6.4.5.2 Rotor impedance Zrs.- The rotor impedance Zrs is the impedance of the rotor with the stator terminals short circuited.

6.4.5.3 Stator impedance Zso.- The stator impedance Zso is the impedance of the stator with the rotor terminals open circuited.

6.4.5.4 Stator impedance Zss.- The stator impedance Zss is the impedance of the stator with the rotor terminals short circuited.

6.4.5.5 Compensating impedance Zco.- The compensating impedance is the impedance of the compensating windings with rotor and stator terminals open circuited.

6.4.6 In-phase component.- The in-phase component is the fundamental frequency component of the output voltage that is in time phase with the energization voltage.

6.4.7 Interaxis error.- Interaxis error is the angular deviation of the null positions for all rotor and stator winding combinations at the 90°, 180°, and 270° rotor angles.

6.4.8 Phase shift.- Phase shift is the difference (expressed in electrical degrees) between the time phase of the fundamental component of the primary voltage and the time phase of the secondary voltage of the resolver at the first position of maximum coupling in the positive direction from resolver zero.

6.4.9 Rated (primary) voltage.- The rated (primary) voltage of a resolver is the line voltage of the resolver's system power supply, i.e., 115 or 26 volts or the voltage range over which the resolver is designed to operate.

6.4.10 Residual voltage.- Residual voltage in a resolver is the actual voltage present at the secondary winding at that position at which the theoretical secondary voltage is zero. This voltage consists of the quadrature component of the energizing frequency and a component which is made up of harmonics of the energizing frequency. Fundamental residual voltage is the quadrature component of the energizing frequency; total residual voltage is the fundamental residual voltage plus harmonics.

6.4.11 Rotor angle.- The rotor angle is the angular mechanical displacement of the shaft from the defined resolver zero position measured in a positive direction.

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6.4.12 Time phase.- The time phase at any point in a resolver system is the phase of the voltage at that point with respect to the phase of the primary voltage of the system. The time phase is measured in electrical degrees.

6.4.13 Transformation ratio.- The transformation ratio is the ratio of the fundamental frequency component of the open circuit secondary voltage and the primary voltage at maximum coupling.

6.4.14 Windings.

6.4.14.1 Compensating winding.- An additional winding which modifies the characteristics of a primary winding, thus correcting imperfections which would otherwise be present in the secondary voltage.

6.4.14.2 Primary winding.- A primary winding is one which receives energizing power from an external source.

6.4.14.3 Secondary winding.- Any winding, other than a compensating winding, which gives an output when a primary winding is energized.

6.5 Qualification.- With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products Lists is Frankford Arsenal, Philadelphia, PA 19137, ATTN: SARFA-MDM and information pertaining to qualification of products may be obtained from that activity.

6.6 International standardization agreements.- Certain provisions of this specification are the subject of international standardization agreements (NATO Electronic Technical Recommendation (NETR) Nos. 10 and 15). When amendment, revision, or cancellation of this specification is proposed which affects or violates the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

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6.7 Performance criteria.- Physical and performance criteria for individual resolver types are specified in the applicable specification sheets (see 3.1) and are based on a close tolerance power supply and on test conditions which are strictly controlled. Under service conditions these values may be modified by the affects of power supply variations and of environmental conditions.

Custodians:
Army - MU
Navy - AS
Air Force - 85

Preparing activity:
Army - MU

Project No. 5990-0300

Review activities:
Army - AV, EL, MI
Navy - OS, TD, EC
Air Force - 11, 17, 79, 80
DSA - ES

User activities:
Army - WC
Navy - MC, CG
Air Force - 14, 19

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